

Why U.S. Flights Run Late

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Data via the United States Department of Transportation

Summary

- Analyzed all 5.8M U.S. domestic flights in 2015 to find when, where, and why flights run late
- 72% of delay minutes come from causes airlines control
- Evening flights are delayed 2.5x as often as morning flights (25.7% vs. 10.1%)
- Airline matters more than airport: delay rates range from 11.3% to 29.7%
- Recommendation: protect morning departures and add turnaround buffer later in the day

Source: U.S. DOT Bureau of Transportation Statistics, 2015 on-time performance (5.8M flights). Delayed = arrived 15+ minutes late.

Two data fixes restored October and cut the delay rate in half

486,165

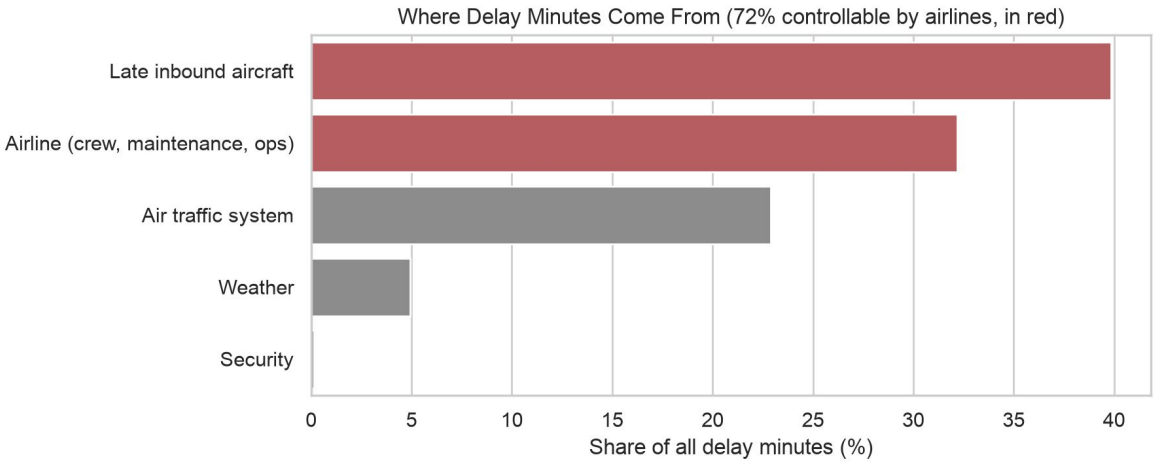
October flights restored. The raw file coded every October airport as a number, so October dropped out of every airport chart. I rebuilt the mapping from September and November routes and recovered 99.97% of them.

36.5% → 18.6%

Share of flights delayed after switching from "any flight 1+ minute late" to the DOT and airline standard of 15+ minutes.

Source: U.S. DOT Bureau of Transportation Statistics, 2015 on-time performance (5.8M flights). Delayed = arrived 15+ minutes late.

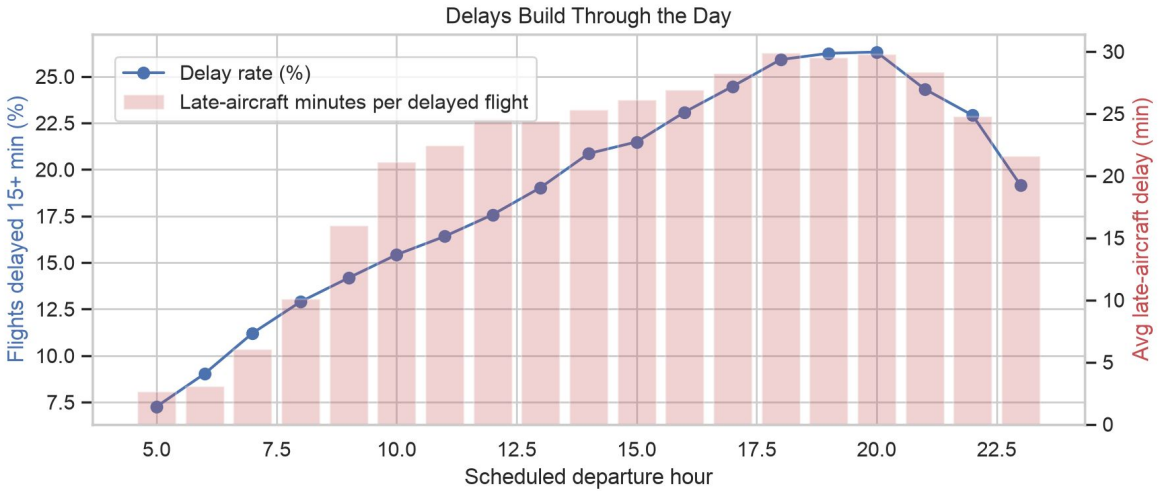
72% of delay minutes come from causes airlines control



- Late inbound aircraft (39.8%) and airline operations (32.2%) make up most delay time
- Weather is 4.9%, though the DOT counts non-extreme weather under the air traffic system
- Spirit is the exception: only 54% of its delay is controllable

Source: U.S. DOT Bureau of Transportation Statistics, 2015 on-time performance (5.8M flights). Delayed = arrived 15+ minutes late.

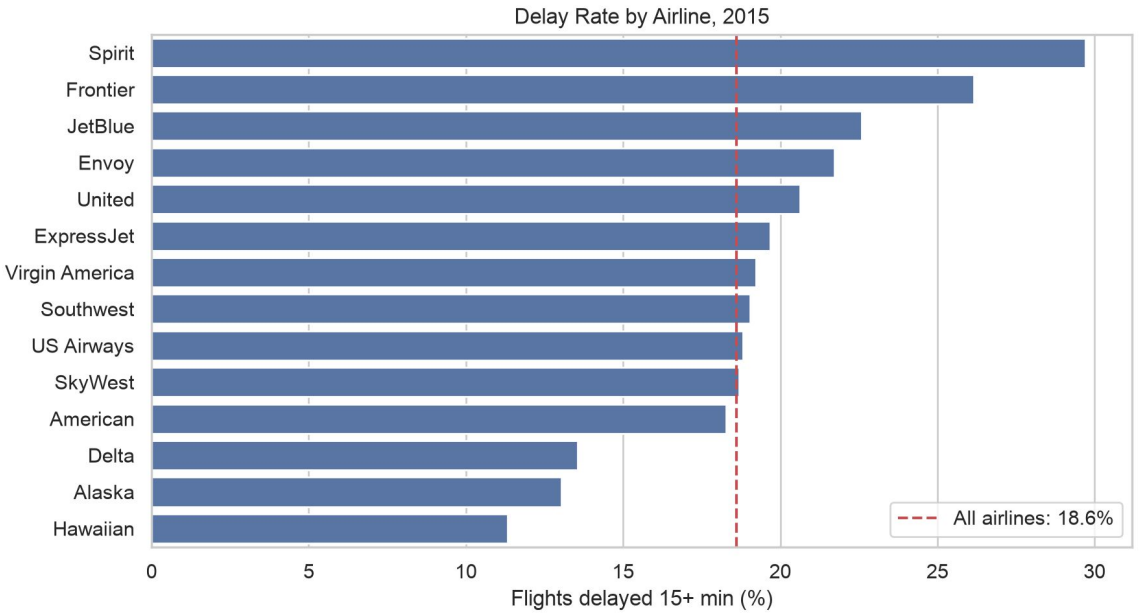
Evening flights are delayed 2.5x as often as morning flights



- 5-8am: 10.1% delayed. 5-8pm: 25.7%
- A delayed 6am flight carries about 3 minutes of late-aircraft delay; a delayed 6pm flight carries about 30
- Each late plane passes its delay to every later flight it flies

Source: U.S. DOT Bureau of Transportation Statistics, 2015 on-time performance (5.8M flights). Delayed = arrived 15+ minutes late.

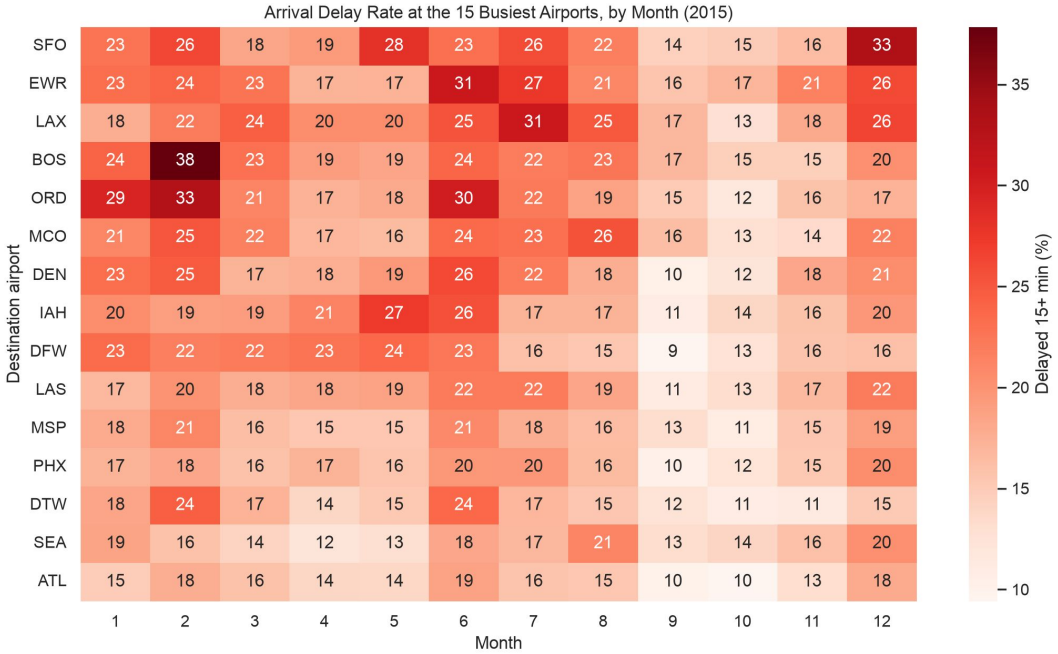
Delay rates vary 2.6x by airline, from Hawaiian (11%) to Spirit (30%)



- Delta flew 876K flights at a 13.6% delay rate, so size isn't the cause
- Among the 15 busiest airports, the range is narrower: 14.7% (ATL) to 22.0% (SFO, EWR)

Source: U.S. DOT Bureau of Transportation Statistics, 2015 on-time performance (5.8M flights). Delayed = arrived 15+ minutes late.

June and winter storms are the worst stretches; October is the calmest



- June: 23.5% of flights delayed, the worst month
- February storms: 38% at Boston, 33% at Chicago O'Hare
- October: 12.4%, the best month and the one my first version was missing

Source: U.S. DOT Bureau of Transportation Statistics, 2015 on-time performance (5.8M flights). Delayed = arrived 15+ minutes late.

Fixing operations is the biggest lever for on-time performance

1

Protect the morning

An on-time first wave keeps late planes from passing delays down the line

2

Add turnaround buffer after noon

Late-aircraft delay is about 10x higher in the evening than in the morning

3

Benchmark Delta and Alaska

Both run large networks at about half of Spirit's delay rate

Limitations: one year (2015) of domestic flights. Next step: re-run the same pipeline on recent BTS data.

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